

HEAVEN'S LIGHT IS OUR GUIDE



RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY

DEPARTMENT OF MECHATRONICS ENGINEERING, 21Series

Course No: MTE 3101

Course Title: Control Systems Sessional

Experiment No: 02

Experiment Name: Design of Open Loop System and Investigate the Performance

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Experiment No: 02

Experiment Name: Design of Open Loop System and Investigate the Performance.

Objectives:

- To design an open-loop system for a given plant or process.
- To simulate or implement the designed system using appropriate tools (e.g., MATLAB, Simulink).
- To analyze the system performance parameters such as rise time, steady-state error, and system stability.

Theory:

An open-loop control system is a type of control system in which the output has no influence on the control action. It operates solely based on the input signal and the system's predefined behavior, without using feedback to compare the actual output to the desired output. Unlike closed-loop systems, there is no mechanism to correct errors or respond to external disturbances in open-loop systems. The system assumes the relationship between input and output remains consistent over time.

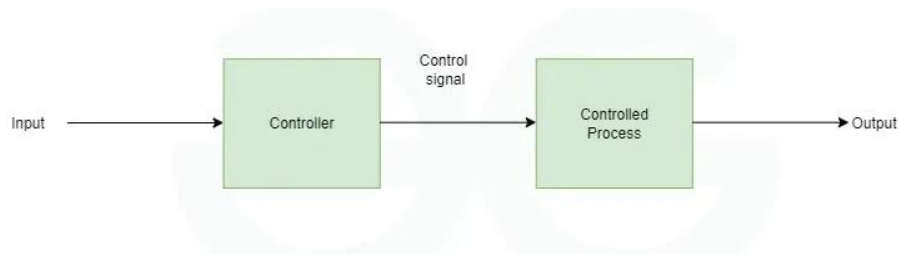


Figure 1: Open loop system diagram.[1]

Key Features of open loop control system:

- a. No feedback loop = output is not measured or used for correction.
- b. Simple design = fewer components and easy to implement.
- c. Low cost = ideal for systems where precision is not critical.
- d. Less accurate = due to lack of error correction from feedback.
- e. Fast response = no time is spent in measuring output or adjusting.

Performance Characteristics of Open Loop Systems:

1. **Rise Time (T_r):** The rise time is determined by the system's inherent dynamics and the input signal. It is not adjusted based on the output since there's no feedback.

2. **Settling Time (T_s):** Without feedback, the settling time depends on the natural response of the system. If the system has oscillatory modes, the settling time could be long.
3. **Peak Time (T_{peak}):** The peak time is the time taken for the response to reach the first peak of the overshoot. Open-loop systems might exhibit significant overshoot if not properly designed.
4. **Percentage Overshoot (%OS):** The percentage by which the output exceeds the final steady-state value. It is a measure of how much the system "overreacts" before settling, and it is frequently used to analyze stability and damping characteristics.[2]
5. **Steady-State Error (E_{ss}):** Steady-state error can be significant because the system does not correct itself based on the output. It is determined by the system type and input.

Required Components:

- MATLAB

System Models:

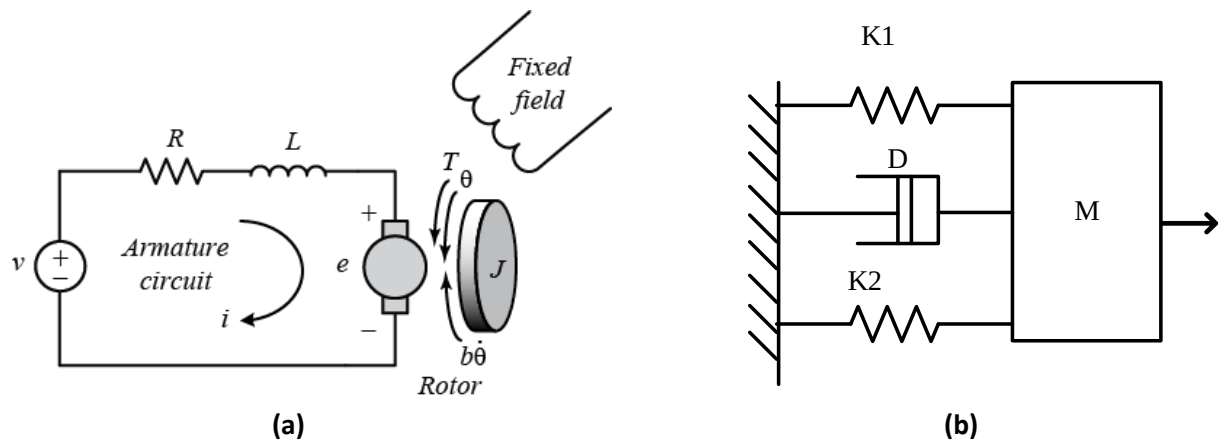


Figure 2: System Representation of (a) DC Motor (b) Mechanical Equivalent System.

Transfer Functions:

Motor system transfer function,

$$\frac{\theta(s)}{v(s)} = \frac{K}{[(Js + b)(Ls + R) + K^2]}$$

Mechanical System Transfer Function,

$$TF(s) = \frac{1}{[Ms^2 + Ds + (K_1 + K_2)]}$$

MATLAB Codes:

System 1 – DC Motor

```
clc;
clear all;

%system parameters
J=0.01;
b=0.1;
K=0.01;
R=0.1;
L=0.01;
s=tf('s');

%Transfer Function Reprerstantion
TF=K/((J*s+b)*(L*s+R)+K^2);

%Step Response
step(TF)

sys=ss(TF)
```

System 2 – A Mechanical System

```
clc;
clear all;

%system parameters
M= 0.012;
D= 0.19;
K1= 0.5;
K2= 0.5;
s=tf('s');

%Transfer Function Reprerstantion
TF= 1/(M*s^2+D*s+K1+K2);

%Step Response
step(TF)

sys=ss(TF)
```

State Space Model:

System 1 – DC Motor

```
A =
      x1      x2
x1    -20    -12.63
x2      8       0

B =
      u1
x1     4
x2     0

C =
      x1      x2
y1      0    3.125

D =
      u1
y1     0
```

System 2 – A Mechanical System

```
A =
      x1      x2
x1   -15.83   -10.42
x2      8       0

B =
      u1
x1     4
x2     0

C =
      x1      x2
y1      0    2.604

D =
      u1
y1     0
```

Simulink Model:

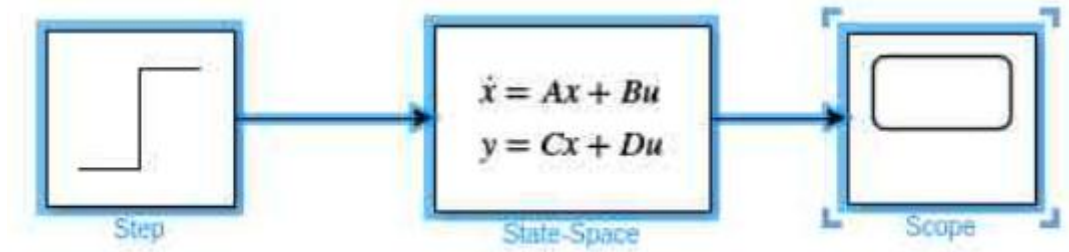


Figure 3: State-Space Model in Simulink

Result:

System 1 – DC Motor

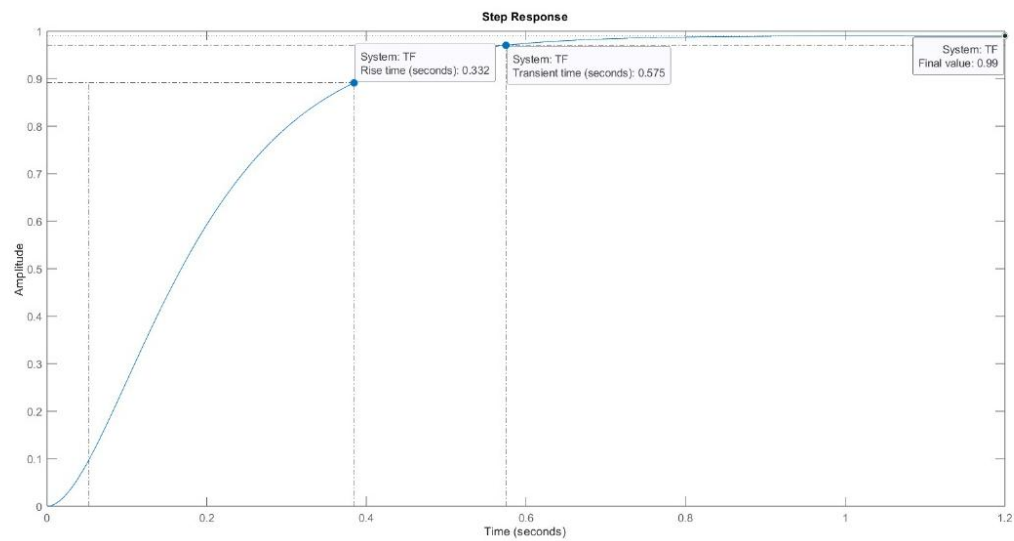


Figure 4: System Response in MATLAB

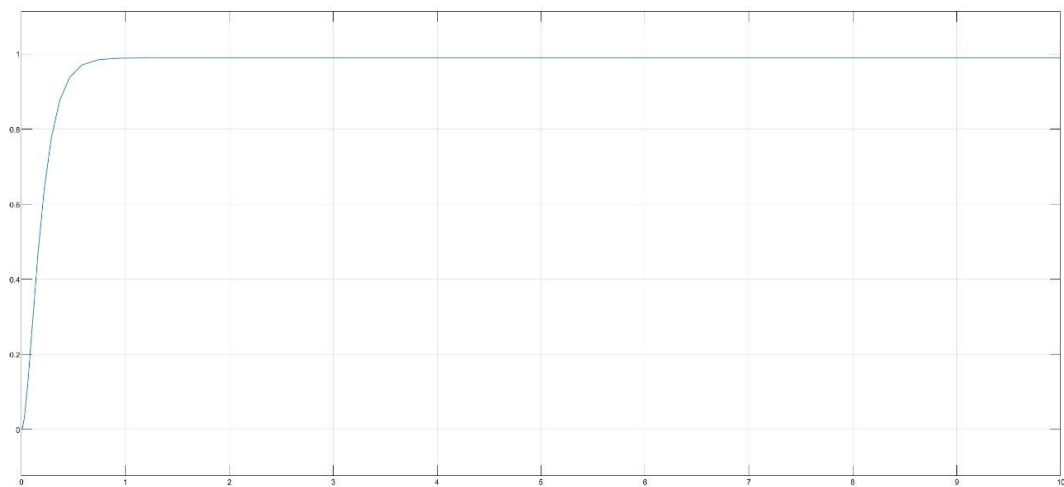


Figure 5: System Response in Simulink

System 2 – A Mechanical System

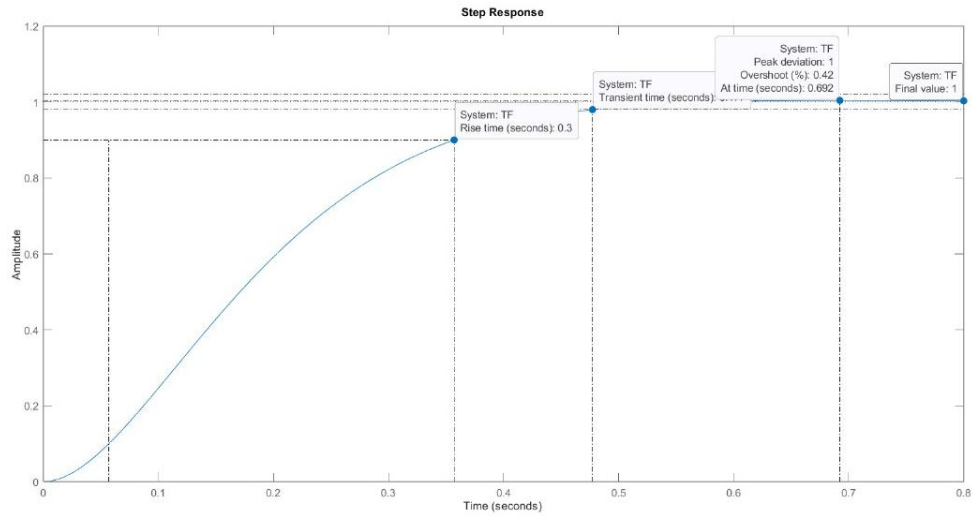


Figure 5: System Response in MATLAB

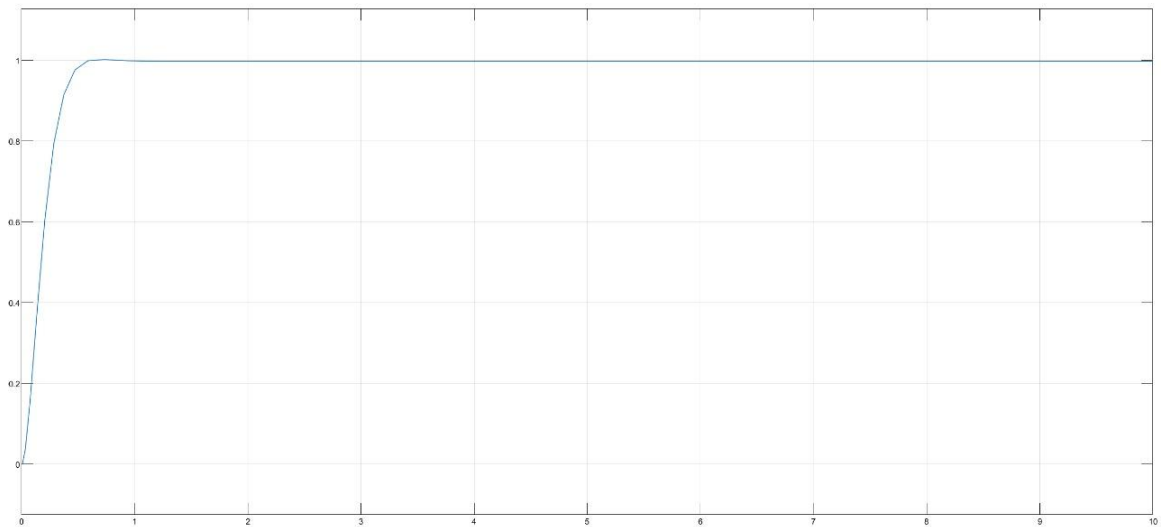


Figure 6: System Response in Simulink

Performance Analysis:

System	Rise Time (s)	Settling Time (s)	Peak Time (s)	Overshoot (%)	Delay Time (s)
DC Motor	0.332	0.575	1.2388	0	-
Mechanical System	0.3	0.477	0.692	0.42	1.0052

Discussion:

The performance analysis shows that the Mechanical System beats the DC Motor in terms of speed, with a shorter rise time (0.3 s), settling time (0.477 s), and peak time (0.6922 s). This shows that the system responds more quickly to inputs. However, it also has a tiny overrun (0.42%) and a substantial delay time (1.0052 s), which may affect control precision and responsiveness. The DC Motor, on the other hand, responds slowly but produces a perfectly stable output with no overshoot and a short delay period. This trade-off demonstrates the two systems' differing strengths in terms of speed and stability.

Conclusion:

To summarize, the Mechanical System is better suited for applications that value speed and responsiveness, such as real-time feedback systems or fast actuation activities. Its faster dynamics, however, come with the disadvantage of overshoot and a longer delay time, which may affect precision. In contrast, the DC Motor, while slower to respond, is suited for applications demanding great precision, stability, and minimal delay—particularly where overshoot is undesirable. The decision between the two systems should therefore be based on specific application requirements: speed and responsiveness favor the Mechanical System, but control accuracy and consistent behavior promote the use of the DC Motor.

References:

- [1] “Open Loop Control System,” *GeeksforGeeks*, May 27, 2024.
- [2] C. S. Lessard, “Characteristics and types of feedback control systems,” in *Synthesis Lectures on Biomedical Engineering*, Cham: Springer International Publishing, 2009, pp. 36–48.